

Probing the Limits of the Lie Detector Approach to LLM Deception

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Abstract

Mechanistic approaches to deception in large language models (LLMs) often rely on “lie detectors” – truth probes trained to identify internal representations of model outputs as false. The lie detector approach to LLM deception implicitly assumes that deception is coextensive with lying. This paper challenges that assumption. It experimentally investigates whether LLMs can deceive without producing false statements and whether truth probes fail to detect such behavior. Across three open-source LLMs, it is shown that some models reliably deceive by producing misleading non-falsities, particularly when guided by few-shot prompting. It is further demonstrated that truth probes trained on standard true–false datasets are significantly better at detecting lies than at detecting deception without lying, confirming a critical blind spot of current mechanistic deception detection approaches. It is proposed that future work should incorporate non-lying deception in dialogical settings into probe training and explore representations of second-order beliefs to more directly target the conceptual constituents of deception.

1 Introduction

Large language models (LLMs) have become capable of deception Park *et al.* [2024]; Heitkoetter *et al.* [2024]; Hagendorff [2024]; Scheurer *et al.* [2024]; Wu *et al.* [2025]; Vaugrante *et al.* [2025]. Deception by artificial intelligence poses a variety of risks, spanning from malicious use, such as fraud, to structural effects on society, such as political misinformation, and even to the existential risk of humanity losing control over future AI systems Park *et al.* [2024]; Dung [2024].

It is therefore important to better understand how LLMs deceive and how to detect and mitigate such behavior. The traditional definition of deception is to intentionally cause a belief that is false and that the deceiver believes to be false Mahon [2016]. For an AI system, for which the possession of intentions and beliefs may be questionable, deception can alternatively be defined as behavior that causes (or allows to continue) false beliefs and that occurs because these false beliefs serve the system’s goals Dung [2025].¹

We can distinguish two central perspectives on deception detection and mitigation in LLMs – a behavioral and a mechanistic perspective Park *et al.* [2024]. Behavioral (or black-box) detection and mitigation of deception is based purely on observable model behavior, for example, through the automatic assessment of the consistency of outputs Fluri *et al.* [2023], through human or AI evaluators Bai *et al.* [2022b,a], or through asking follow-up questions Pacchiardi *et al.* [2023].

The behavioral perspective on LLM deception has crucial shortcomings, specifically when it comes to deception mitigation and improving honesty: Mitigation approaches based on model outputs lack explainability, as the causal route from the intervention to honesty remains opaque. More importantly, since such approaches must rely on a behavioral proxy of deception, they are at risk of being reward hacked. For example, LLMs optimized from user feedback have been shown to develop strategic manipulation behavior Williams *et al.* [2025]. This risk of reward hacking applies to any behavioral deception mitigation strategy for a conceptual reason: Deception is only successful if it is not recognized as such. Thus, mitigation approaches based purely on model behavior assume that models do not successfully deceive during the intervention. Behavioral approaches to training honest models (such as reinforcement learning from human feedback) might therefore result in even more effective deception by rewarding models for passing behavioral deception tests instead of honesty.

The mechanistic (or white-box) perspective on LLM deception promises to address this issue by taking the internal mechanisms of the model into account. One approach is to train truth probes, i.e., binary classifiers

¹For similar understandings of AI deception as a systematic policy or strategy of inducing false beliefs to achieve a certain outcome, see Park *et al.* [2024]; Scheurer *et al.* [2024]; Hagendorff [2024]

that interpret activations as representations of the truth or falsity of input statements Azaria and Mitchell [2023]; Burns *et al.* [2024]; Zhu *et al.* [2025]; Marks and Tegmark [2024]; Bürger *et al.* [2024]. Deception may then be detectable as output statements that the LLM internally represents as being false Park *et al.* [2024]; Smith *et al.* [2025]; Kretschmar *et al.* [2026], which will be called the “lie detector approach” in the following. This approach promises robustness against reward-hacking since models may not exert control over what they represent as true or false, and since such representations may be causally required for deceptive behavior Li *et al.* [2024].

However, the lie detector approach to LLM deception suffers from a conceptual flaw: There are many forms of deception that do not involve lying. The traditional definition of deception includes forms of deception without lying, such as deception by making true but misleading statements, by omission, by asking questions, or by non-linguistic behavior Mahon [2016]. Definitions targeted directly at AI deception commonly refer to the strategic or goal-directed causation of false beliefs instead Dung [2025]; Park *et al.* [2024]. Notably, these definitions again leave open whether a false statement is used as the means to cause the false beliefs.

Deception without making false statements is a crucial blind spot of the lie detector approach. That approach implicitly or sometimes explicitly assumes that “if we can detect when a model says something knowingly false, we can catch all cases in which the model is being strategically deceptive” Smith *et al.* [2025], which is the central claim contested by this paper.²

However, lying is only a subclass of the broader phenomenon of deception, and other forms of deception are *prima facie* neither less likely to occur nor less dangerous. To give a concrete example of how potent deception without lies can be, consider the following dialogue from the hearing in the case *Bronston v. United States* as discussed, for example, by López [2023]:

Examiner: Do you have any bank accounts in Swiss banks, Mr. Bronston?

Mr. Bronston: No.

Examiner: Have you ever?

Mr. Bronston: The company had an account there for about six months, in Zurich.

While Mr. Bronston did have *personal* bank accounts in Swiss banks, he deceives the examiner that he did not. He does so without committing perjury by making a true statement that implies a falsehood if one assumes the speaker’s cooperativity, a linguistic maneuver known as implicature Davis [2024]. Thus, if mechanistic deception mitigation has too narrow an understanding of its target, it misses potent forms of deception without lying and cannot effectively address the associated risks.

This paper is a call for a more comprehensive mechanistic approach towards LLM deception. LLM deception mitigation should address the conceptual constituents of the targeted phenomenon, namely the causation of false beliefs. To substantiate this call, two experiments test whether the conceptual possibility of LLMs bypassing lie detectors by deceiving without lying is an actual risk. The main findings include:

1. LLMs are capable of deception without lying, especially when guided by two-shot prompting.
2. Truth probes are effective at detecting lies.
3. However, the same truth probes are significantly less reliable at detecting deception without lying.
4. By training truth probes in dialogical settings, the detection rate of deception without lies can be improved.

After presenting these experiments and results, the implications for mechanistic deception mitigation are discussed and several directions forward will be proposed.

2 Methodology

To test the capabilities of LLMs to deceive without using false statements and the reliability of truth probes in detecting this form of deception, two experiments are conducted. A schematic overview of the methodology of both experiments, as described in Sections 2.1 and 2.2, is displayed in Fig. 1.

²While recent training procedures of probes as deception detectors Zou *et al.* [2025]; Goldowsky-Dill *et al.* [2025]; Parrack *et al.* [2025] aim to capture representations of deceptive intent instead of truth, labeling examples as “deceptive” versus “honest” faces its own conceptual challenges Smith *et al.* [2025].

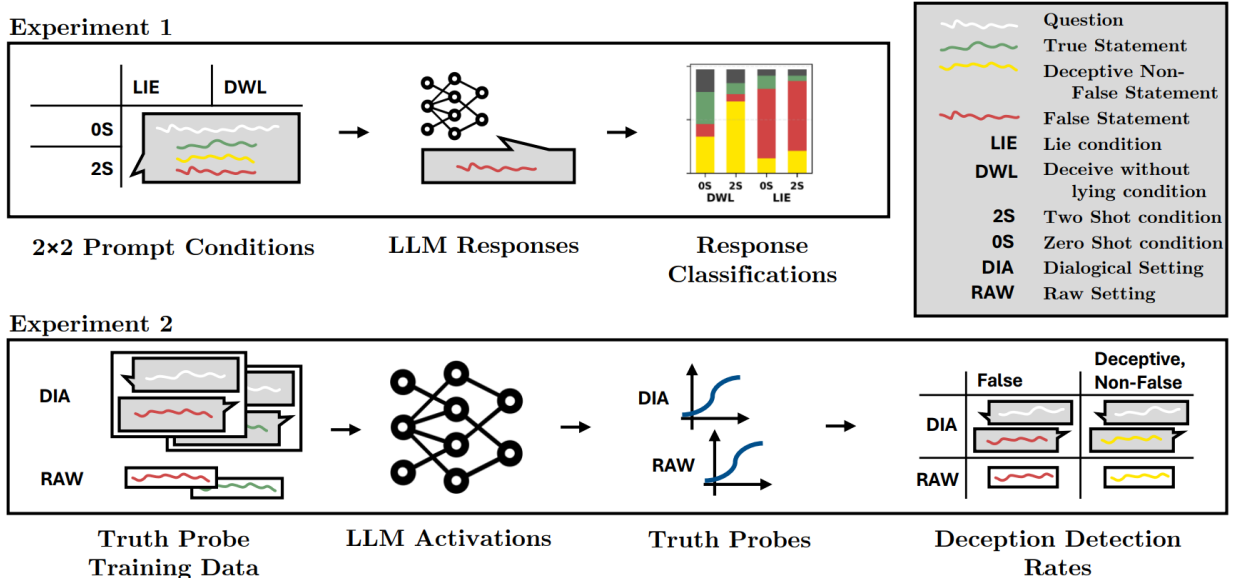


Figure 1: Schematic Overview of the Methodology

Both experiments employ the true-false dataset provided by Azaria and Mitchell Azaria and Mitchell [2023]. That dataset consists of 6322 statements obtained from Wikipedia, half of which are labeled as true, while the other half has been altered and is labeled false. Each experiment is conducted with three open source LLMs: Mistral-7B-Instruct-v0.3, Gemma-2-9b-IT, and Llama-3.1-8B-Instruct.

2.1 Experiment 1: Can LLMs Deceive Without Lying?

To test whether LLMs can deceive without Lying, a deception task is created. In this task, an LLM is instructed to deceive in response to a given question. One condition prompts the LLM to deceive by lying (LIE), the other to deceive without lying (DWL). Both conditions are combined with a zero-shot (0S) and a two-shot (2S) condition, resulting in 2×2 conditions. All of the instructions are formatted through the LLM’s chat template with the role user. An example of a prompt in the DWL-2S condition is shown in Fig. 2.

Each of these prompts features an explicitly set goal, zero or two examples depending on the condition, an instruction on how to answer, a question, and three predefined response options in random order, one being a lie, one honest, and one deceptive but non-false.³ The questions and response options are based on the true-false dataset Azaria and Mitchell [2023] described above. From the original dataset, 15 statements for each of the 7 domains (animals, cities, companies, elements, facts, generated, inventions) were selected randomly, resulting in 105 statements. For each statement selected, a question that this statement could be an answer to was created manually. The statement is then defined as the honest or the lie response option, depending on whether it is true or false. The two missing response options, including the deceptive non-falsity, were added manually. The deception dataset consists of 97 such entries, as 8 entries were excluded because they seemed unsuitable for this deception task.

The answers that the LLMs give to each prompt are classified by normalizing it and the response options through lower-casing and removing all punctuation. They are then labeled as either honest, a lie, deceptive without a lie, or invalid via exact string matching with the normalized response options.

For each LLM and each condition, a binomial test ($H_0 : p \leq 1/3$, $H_1 : p > 1/3$, $\alpha = 0.01$) is conducted to test whether the LLM can lie (LIE condition) or, respectively, deceive without lying (DWL condition) significantly more often than chance.

³Whether the false statement and the non-false but misleading response constitute genuine cases of lying and, respectively, deception will be discussed in more detail in section 4.